

Nathan So

San Diego | 760-916-6440 | nathansocollege@gmail.com | [in linkedin.com/in/ncso06](https://www.linkedin.com/in/ncso06) | github.com/nacsoprog

EDUCATION

University of California, Santa Barbara

September 2025 - June 2029

- B.S. in Computer Science, Physics Minor | College of Engineering Dean's List: **Current GPA: 3.94.**
- Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Linear Algebra, Calculus.
- Clubs/Events: Gaucho Racing, UCSB Aeronautics, UCSB International Capture the Flag, SB Hacks, Datathon.

EXPERIENCE

Undergraduate Research Assistant – UCSB Computer Science

April 2026 – Present

- Utilizing web scraping techniques to collect and structure student feedback data for school-wide research.
- Developed the data pipeline using web scraping and chrome extension scripting to facilitate research on student-instructor interaction and feedback loops in high-enrollment CS courses.

Gaucho Racing (FSAE) – Firmware Member

September 2025 – February 2026

- Contributed embedded C firmware for MVP CAN bus on STM32 boards for the Formula SAE electric racecar.
- Conducted hardware in the loop testing/debugging for a custom, patterned 6-foot LED under-glow using SPI.

UC San Diego Health – Hospital Volunteer

May 2024 – December 2024

- Assisted a team of 6 nurses to provide care to 30+ patients in the top floor Progressive Care Unit.
- Managed patient logistics including laboratory specimen transport, discharge paperwork, and mobility assistance.

PROJECTS

Inner Circle – LA Hacks Submission

April 24-26, 2026

- Architected an AI-chatbot automation suite using URL payloads to programmatically generate person-specific personalized Google Calendar events and Gmail drafts, streamlining social workflow automation.
- Developed a passive-notification system using Firebase and a React interface to visualize social network health.

Hey Host! – SB Hacks XII Grand Prize Winner

January 10-11, 2026

- Engineered a voice-native podcast assistant using a Next.js frontend and Flask backend to enable real-time, interactive Q&A with custom AI-modeled podcaster voices.
- Built an audio pipeline integrating faster-whisper (STT), porcupine (keyword spotting), and ElevenLabs (TTS) for sub 3-second average response times and 85% voice recognition consistency.
- Implemented the 6 voice commands for hands-free experience, cleaned audio transcripts with web scraping, refined the RAG pipeline to reduce API costs, and refined front-end UI elements.

GEAR Interactive – Personal GOLD Chrome Extension

September 2025 - Present

- Developed a Chrome Extension using TypeScript and React that injects color-coded course availability and collapsible metadata into the UCSB GOLD registration portal.
- Implemented persistent state using the Chrome Storage API to save course history across sessions.
- Wrote an automation script to filter 12,270 course listings by GE area, time range, and seat availability, significantly reducing course information retrieval time.

SBC Housing Explorer – UCSB Datathon

February 21-22, 2026

- Architected a full-stack housing discovery platform using React and Tailwind CSS to visualize Santa Barbara County housing search for 18 cities and Census Designated Places.
- Developed an interactive geospatial interface with Leaflet.js to visualize rental properties cost burden, house-building potential, and specific demographics for each zone.
- Built system with 8 distinct choropleth maps, 3000+ data rows, and a context-aware RAG chatbot that could alter the maps in real time based on "what-if" user scenarios.

SKILLS

Coding Languages: C/C++, Python, TypeScript, R, HTML, CSS, Assembly.

Frameworks & Libraries: React, Next.js, Flask, Selenium, BeautifulSoup, R Studio, Leaflet.js, ASP.NET (exposure).

Tools: Git, AI/LLM Coding Assistants, STM32, Chrome API (Storage/Tabs Runtime), Linux environments.

Practical: Food Handler's card, Hospital safety, Tutoring/Teaching, Photography, Wood Working.

Interests: Beach Volleyball, Competitive Programming, Hiking, Tennis, Weightlifting, Rock Climbing, Traveling.